

Defective Analysis: Arithmetic Ladder from 0 to $\mathbb{C}$

Typed reconstruction derivation of the standard algebraic structures

Terminal statement

In matured Defective Analysis, number systems are not treated as untyped universal objects. They are typed repair objects: each algebraic structure appears when the prior chart has a residual that cannot be reconstructed by the currently admitted operators, projectors, quotients, and retained fibers.

$$0 \longrightarrow \mathbb{N} \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Q} \longrightarrow \mathbb{R} \longrightarrow \mathbb{C}.$$

Equivalently,

$$\boxed{\mathbb{C} = \text{Adm}_{\text{DA}}(0, 1; \theta^\pm, \alpha^\pm, \lambda, \rho)}$$

where

$$\begin{aligned} \theta &= \text{derivation/access/successor axis}, & \alpha &= \text{integration/reconstruction/gluing axis}, \\ \lambda &= \text{limit-completion defect}, & \rho &= \text{polar/phase-closure defect}. \end{aligned}$$

The compressed ladder is

$$\boxed{\mathbb{N} = (\theta^+, \alpha^+), \quad \mathbb{Z} = (\theta^\pm, \alpha^+), \quad \mathbb{Q} = (\theta^\pm, \alpha^\pm), \quad \mathbb{R} = (\theta^\pm, \alpha^\pm, \lambda), \quad \mathbb{C} = (\theta^\pm, \alpha^\pm, \lambda, \rho).}$$

DA construction rule

For a typed carrier X in a declared projector sector P , DA exposes a quotient layer and a retained reconstruction fiber:

$$\begin{aligned} D_{\theta, P}X &= (C_X, K_X), \\ I_{\alpha, P}(C_X, K_X) &\sim_E X. \end{aligned}$$

Here C_X is the exposed quotient/COTG layer and K_X is the retained residual fiber needed for faithful reconstruction. A new algebraic structure is admitted when the flat reconstruction

$$I_{\alpha, P}^0(C_X) = I_{\alpha, P}(C_X, 0)$$

fails. The next structure is the minimal admissible repair of the residual:

$$X_{n+1} = \text{Repair}_{\text{DA}}(X_n; \text{res}(X_n)).$$

0. Zero class

The DA zero class is not the empty set. It is the equivalence class of no exposed distinction:

$$0 := [X]_{D_\theta X=0} = \{X : D_\theta X \text{ exposes no distinguishable sector}\}.$$

It is a collapsed chart: no successor, no comparison, no orientation, no ratio, no limit, no phase. Its defect is that it cannot distinguish zero from non-instantiation. The first repair is a witnessed unit:

$$0 \rightsquigarrow 1.$$

The seed $(0, 1)$ is the minimal distinction generator.

1. Natural numbers: forward generative closure

Admit the forward successor operation

$$S_\theta(n) = n + 1.$$

Then

$$\mathbb{N} = \{0, 1, S_\theta(1), S_\theta^2(1), \dots\} = \{0, 1, 2, 3, \dots\}.$$

In DA notation,

$$\boxed{\mathbb{N} = \text{cl}_{\theta^+, \alpha^+} \{0, 1\}.}$$

The induced algebraic structure is the commutative semiring

$$(\mathbb{N}, +, \cdot, 0, 1),$$

where addition is repeated successor and multiplication is repeated addition.

Residual:

$$n + ? = 0$$

has no solution in $\mathbb{N}$ for $n > 0$. Therefore $\mathbb{N}$ is not closed under additive reversal.

Known unknown: whether every forward construction should be typed as θ^+ , or whether some finite accumulations belong to α^+ .

Unknown unknown: whether DA should distinguish counting successor from structural generation as separate positive clocks.

2. Integers: additive inverse closure

The defect of $\mathbb{N}$ is missing additive inverse. DA repairs this by admitting θ^- , the inverse successor direction.

Construct

$$\mathbb{Z} = \mathbb{N} \times \mathbb{N} / \sim$$

with

$$(a, b) \sim (c, d) \iff a + d = b + c.$$

Interpret

$$[(a, b)] = a - b.$$

Then

$$0 = [a, a], \quad n = [n, 0], \quad -n = [0, n].$$

Thus

$$\boxed{\mathbb{Z} = \text{Repair}_{\theta^-}(\mathbb{N}).}$$

The induced structure is the commutative ring

$$(\mathbb{Z}, +, \cdot, 0, 1).$$

Additive residuals are repaired:

$$n + (-n) = 0.$$

But multiplicative residuals remain:

$$n \cdot ? = 1$$

has no integer solution for most $n \neq \pm 1$.

Known unknown: whether $\mathbb{Z}$ is the first true defect-symmetric structure or merely the additive group completion of $\mathbb{N}$.

Unknown unknown: whether DA has a more primitive notion of signed curvature before additive inverse is formalized.

3. Rationals: multiplicative inverse closure

The defect of $\mathbb{Z}$ is missing division. DA repairs this by admitting α^- , the inverse gluing operation for multiplicative accumulation.

Construct

$$\mathbb{Q} = \mathbb{Z} \times (\mathbb{Z} \setminus \{0\}) / \sim$$

with

$$(a, b) \sim (c, d) \iff ad = bc.$$

Interpret

$$[(a, b)] = \frac{a}{b}.$$

Thus

$$\boxed{\mathbb{Q} = \text{Repair}_{\alpha^-}(\mathbb{Z}).}$$

The induced structure is the field

$$(\mathbb{Q}, +, \cdot, 0, 1).$$

Every nonzero element has a multiplicative inverse:

$$q \neq 0 \implies q^{-1} \in \mathbb{Q}.$$

But $\mathbb{Q}$ has limit gaps. For example,

$$x^2 = 2$$

has no rational solution. So $\mathbb{Q}$ is algebraically stronger than $\mathbb{Z}$, but geometrically defective: it cannot reconstruct all limit-stable magnitudes.

Known unknown: whether α^- should be interpreted strictly as division or more generally as inverse gluing/history decomposition.

Unknown unknown: whether irrationality in DA is best framed as missing point, missing limit fiber, or failed finite reconstruction certificate.

4. Reals: limit-completion closure

The defect of $\mathbb{Q}$ is incompleteness under convergent approximation. DA repairs this by admitting λ , the limit-completion defect.

Construct $\mathbb{R}$ as Cauchy completion:

$$\mathbb{R} = \text{Cauchy}(\mathbb{Q}) / \sim,$$

where

$$(a_n) \sim (b_n) \iff \lim_{n \rightarrow \infty} (a_n - b_n) = 0.$$

Thus

$$\mathbb{R} = \text{Repair}_\lambda(\mathbb{Q}).$$

The induced structure is the complete ordered field

$$(\mathbb{R}, +, \cdot, <, 0, 1).$$

Magnitude gaps are repaired: every Cauchy-stable rational approximation receives a real endpoint. But $\mathbb{R}$ still has a phase defect:

$$x^2 + 1 = 0$$

has no real solution. Therefore $\mathbb{R}$ is line-complete, not rotation-complete.

Known unknown: whether λ should be treated as a clock, a completion operator, or a residual-fiber admission rule.

Unknown unknown: whether DA should distinguish metric completion, order completion, and measure completion as separate λ -sectors.

5. Complex numbers: polar/phase closure

The defect of $\mathbb{R}$ is missing polar closure. DA repairs this by admitting ρ , the phase/rotation defect.

Construct

$$\mathbb{C} = \mathbb{R}[x]/(x^2 + 1).$$

Let

$$i = [x], \quad i^2 = -1.$$

Every complex number is

$$z = a + bi, \text{ where } a, b \in \mathbb{R}.$$

Equivalently,

$$\mathbb{C} \cong \mathbb{R}^2$$

with multiplication

$$(a, b)(c, d) = (ac - bd, ad + bc).$$

Thus

$$\mathbb{C} = \text{Repair}_\rho(\mathbb{R}).$$

In DA notation,

$$\mathbb{C} = \text{cl}_{\theta^\pm, \alpha^\pm, \lambda, \rho}\{0, 1\}.$$

The induced structure is an algebraically closed field:

$$(\mathbb{C}, +, \cdot, 0, 1).$$

The real residual

$$x^2 + 1 = 0$$

is repaired by admitting

$$i = \sqrt{-1}.$$

In DA language, i is not merely imaginary. It is the minimal retained phase fiber required to reconstruct rotation and polar closure from a line-complete field.

Known unknown: whether ρ should be identified with phase, polarity, orientation curvature, or exponential closure.

Unknown unknown: whether the next DA repair after $\mathbb{C}$ is naturally quaternionic, operator-algebraic, categorical, or physical rather than purely algebraic.

Compressed derivation

$$0 \xrightarrow{1, \theta^+} \mathbb{N} \xrightarrow{\theta^-} \mathbb{Z} \xrightarrow{\alpha^-} \mathbb{Q} \xrightarrow{\lambda} \mathbb{R} \xrightarrow{\rho} \mathbb{C}$$

Meaning:

- 0 = no exposed distinction,
- 1 = first witnessed distinction,
- $\mathbb{N}$ = forward generative closure,
- $\mathbb{Z}$ = additive inverse closure,
- $\mathbb{Q}$ = multiplicative inverse closure,
- $\mathbb{R}$ = limit/magnitude completion,
- $\mathbb{C}$ = polar/phase completion.

Terminal DA compression:

$$\mathbb{C} = (0, 1) + \theta^\pm + \alpha^\pm + \lambda + \rho.$$

Equivalently:

$\mathbb{C}$ = minimal admissible arithmetic WDS generated by $(0, 1)$
after closure under successor, reversal, ratio, limit, and phase.

At each stage, the quotient kills only the defect being repaired, while the retained fiber preserves the reconstruction data needed for admissibility in the previous chart.